JAIN DEEMED-TO-BE UNIVERSITY | SCHOOL OF COMPUTER SCIENCE AND IT

DEPARTMENT OF COMPUTER SCIENCE AND IT MCA

PROGRAMME

ACADEMIC YEAR: 2026-2027

ACTIVITY-2

PREDICTIVE ANALYSIS AND DATA VISUALIZATION

2SMCAAI302

SEM: 3 SPECIALIZATION: MCA-AIML

Topic: Predicting Library Usage Using Predictive Analytics

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Assistant Prof., Dept. of CS &IT.

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Certificate

This is to certify that **Aman Kumawat (25MCAR0023)** has satisfactorily completed the **Activity-2** entitled **"Predicting Library Usage Using Predictive Analytics"**, prescribed by **Jain (Deemed-to-be University)** as part of the requirements for the **Semester III, Master of Computer Applications (M.C.A.)** programme during the academic year **2026-2027.**

The work presented in this report is completed as part of the prescribed academic activity and has been carried out under the guidance and supervision of the concerned faculty.

Date: 28/10/2026

Aman

Signature of Student

Signature of Faculty Incharge

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INTRODUCTION

University libraries provide students with books, journals, reference materials and other learning resources. Every day, libraries generate transaction data such as borrowing dates, return dates, student details and department information.

This project focuses on analysing university library borrowing data using Python. The main purpose is to understand how students use library.

The project includes data cleaning, data analysis, feature creation and data visualisation. The dashboard helps compare library usage based on department, gender, academic year and time.

Purpose of the Project

The main purpose of this project is to convert raw library transaction data into meaningful information, using data analysis and visualisation.

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DATASET AND PROBLEM STATEMENT

Dataset Description -

The project uses a University library transaction dataset.

- Membership Number
- Borrow Date
- Return Date
- Borrow Duration
- Date of Birth
- Gender.
- Department Code
- Academic Year
- Cumulative GPA
- Total Credit Cumulative

Data Cleaning

Befor analysis, the dataset is cleaned to improve the quality of the data.

- Converting dates into proper date formate
- Converting numerical values into numeric format.
- Coverecting the decimal format in GPA values.
- Removing extra spaces from column names.
- Checking of missing values.
- Handling invalid values during conversion.

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Problem Statement

The project aims to answer questions such as:

- which department uses the library the most?
- which gender group has more library transactions?
- How long do students keep borrowed resources?
- Which months have higher library usage?
- How does library usage change across academic years?
- Is there a relationship between GPA and library usage?

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DASBOARD AND VISUALIZATIONS

1. GPA vs Borrowing

A scatter plot compares students' GPA with the number of resources they borrow.

2. Department and Gender

A bar chart compares library transactions between departments and gender groups.

3. Correlation Heatmap

The heatmap shows relationships between GPA, credits, borrowing duration and frequent user status.

4. Monthly Borrowing

A time-series chart shows how library borrowing change from month to month.

5. Department Usage

A pie chart shows the percentage of library transactions contributed by each department

6. Borrow Duration

A histogram shows how long students usually keep borrowed resources.

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7. Frequent Vs Non-Frequent Users

A histogram compares CGPA distributions between frequent and non-frequent library users.

8. Gender Usage

A bar chart shows the total number of transactions by gender.

9. Academic Year Usage

A bar chart compares library transactions by gender across academic years.

10/26

University Library Analytics Dashboard

Interactive insights from borrowing records

Theme: Dark Light Slate Emerald Sunset

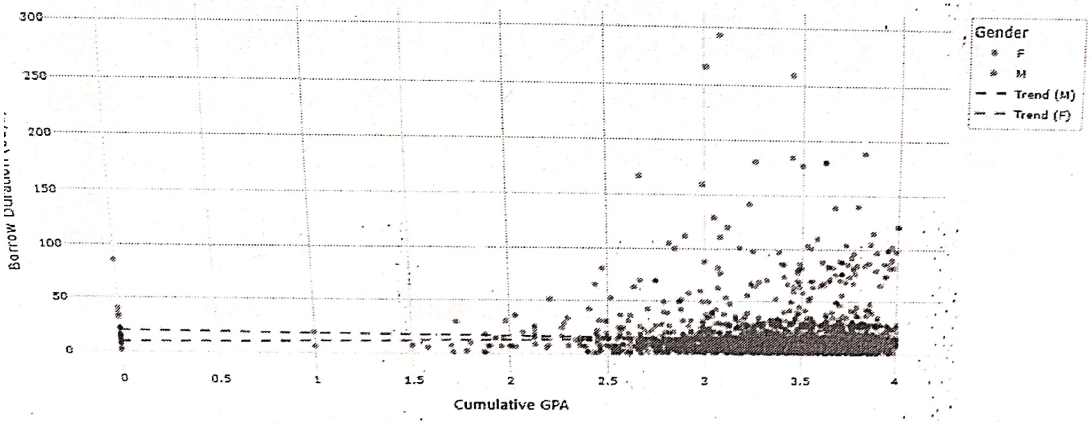
Academic Year: 2014 – 2024 (2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024)

Department: All Departments

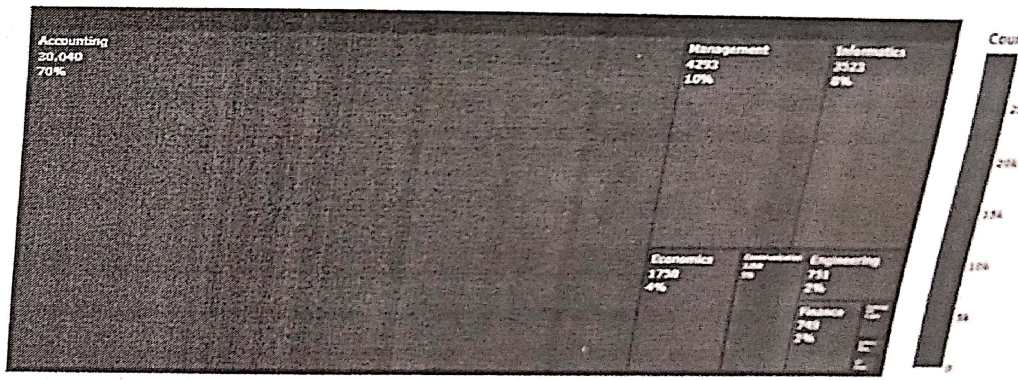
Gender: All

42,701	11,580	7,557	3.40	14.7d
TOTAL TRANSACTIONS	UNIQUE MEMBERS	UNIQUE BOOKS	AVG GPA	AVG BORROW DURATION

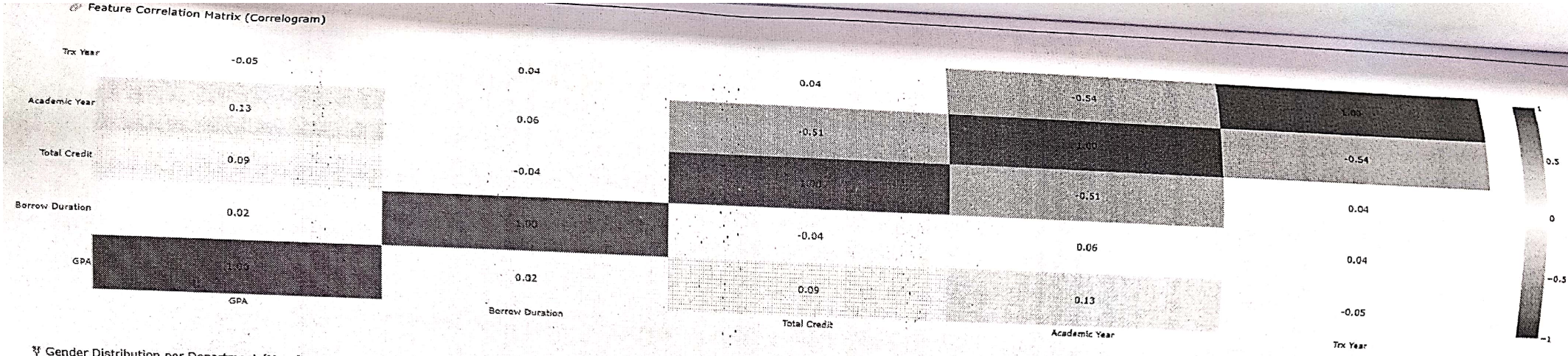
GPA vs Borrow Duration (Scatter Plot)

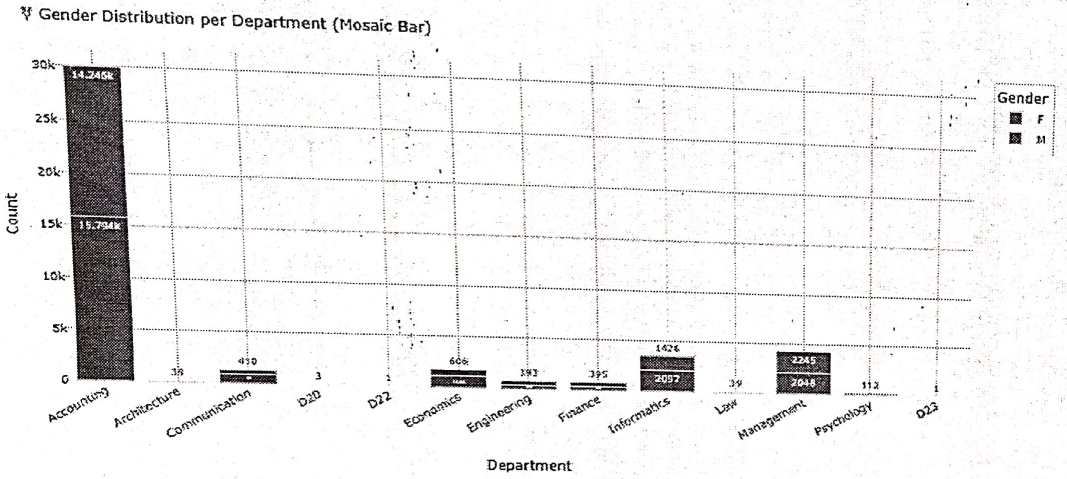


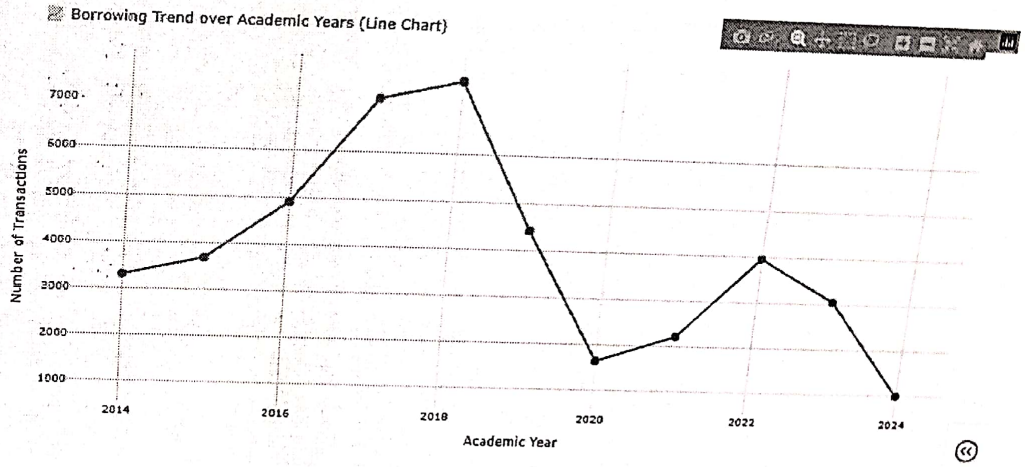
Books Borrowed per Department (Treemap)



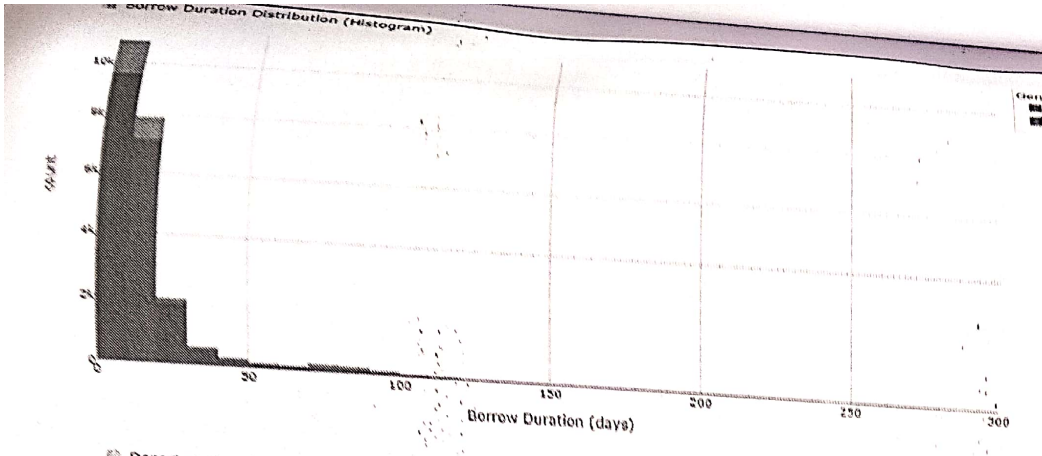
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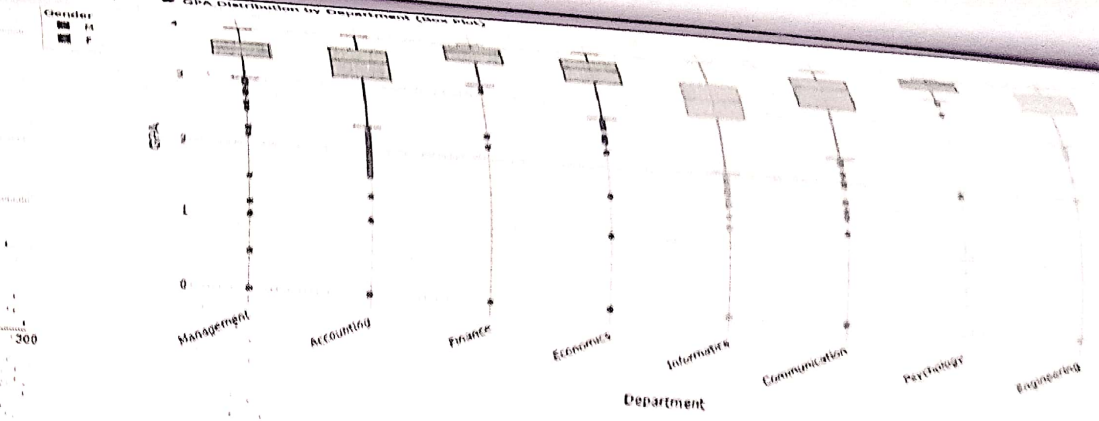




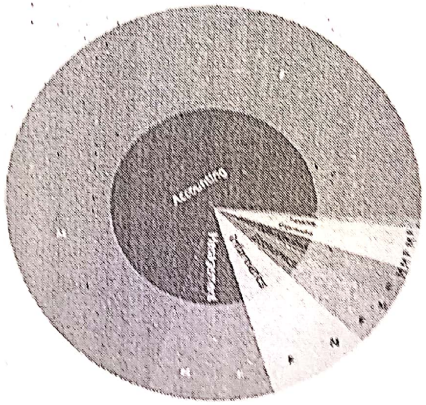


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Department → Gender Hierarchy (Sunburst Chart)



Actionable Insights

Most Active Department

Accounting leads in library usage among filtered records.

Dominant Borrower Gender

Female students represent the majority of borrowing transactions.

Highest Average GPA

Students in D23 have the highest avg GPA (3.92) among filtered departments.

Borrow Duration Insight

Average borrow duration is 14.7 days. High outliers may indicate unreturned books or system recording issues.

Correlation Insight

The correlogram reveals that GPA and borrow duration have a weak correlation, suggesting reading habits alone don't strongly predict academic performance.

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Conclusion

The University Library Usage Dashboard provides a simple way to understand library borrowing behaviour.

The project converts raw transaction data into useful information through data cleaning, analysis and visualization.

The dashboard helps identify differences in usage across departments, gender groups and academic years. It also provides information about borrowing duration and possible relationships between library usage and academic performance.

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